HeatAtlas: Forecast Heat. Prioritise Relief.

Autonomous Extreme Heat Relief Engine & Hyper-Local Spatial Triage for Outdoor Workers & NGOs

OFFICIAL HACKATHON TEAM

Team Strawhats

TEAM CODE: 627JUH

CHOSEN TRACK — THEME 2

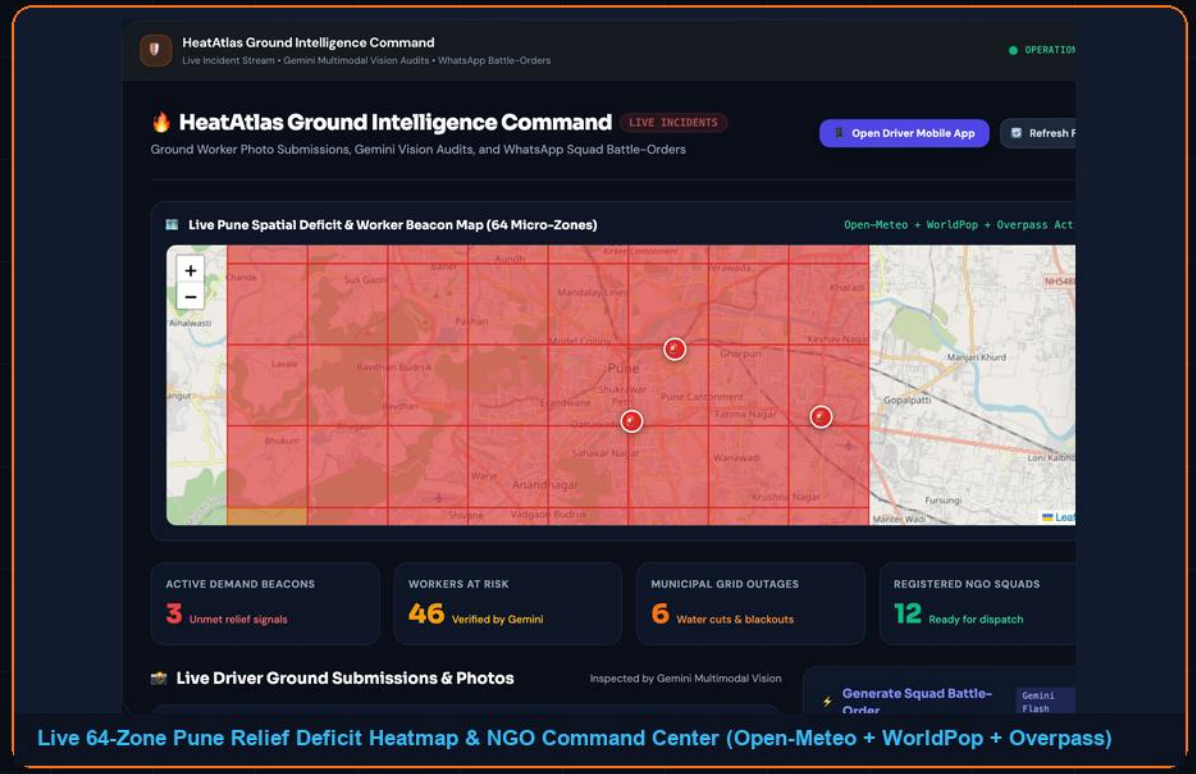
Climate, Resource Infrastructure & Global Supply Resilience (Autonomous Emergency Response Swarm & Urban Resource Controller).

CORE AI + SPATIAL INNOVATION

Zero-Hallucination Split: Deterministic Python computes 64-zone Heat Exposure (Eh), Vulnerability (Vh) & Relief Deficit (Rd); Gemini 2.5 Flash drives NL intake, tool calls & vision audits.

100% LIVE DEPLOYMENT & DATA

Live Open-Meteo 72h Forecast + WorldPop v2 Demographics + OSM Overpass Water/Clinic Nodes. Hosted at https://heatatlas.vercel.app/



Live 64-Zone Pune Relief Deficit Heatmap & NGO Command Center (Open-Meteo + WorldPop + Overpass)

64 Zones — Pune 8x8 Micro-Grid

Zero Mock — 3 Live Public APIs

< 60 Sec — WhatsApp Battle Brief

44.2°C — Peak Corridor Heat

11/11 Pass — Pytest Spec Suite

12 Repos — Integrated Ecosystem

Why City-Wide Weather Alerts Fail During 44°C Urban Heatwaves

When Indian cities breach 44°C, thermal exposure and hydration failure happen street-by-street, not city-wide.

01 • THE SINGLE-NUMBER WEATHER BLINDSPOT

Standard weather apps show one city-wide number ('40°C Pune'). Yet a delivery rider on unshaded Swargate asphalt experiences 44.2°C apparent heat with zero working water taps, while leafy neighborhoods nearby have ample shade and clinics.

Target Audience: 10M+ Outdoor Gig & Transit Workers + NGO Relief Teams

02 • 10M+ OUTDOOR GIG & TRANSIT WORKERS EXPOSED

Delivery riders, street vendors, traffic police, and daily-wage construction crews spend 8–10 continuous afternoon hours outdoors. They lack a fast, low-friction way to locate the nearest verified working drinking water tap or shaded rest stop.

Impact: Severe dehydration and preventable occupational heat exhaustion.

03 • CASCADING WATER & GRID INFRASTRUCTURE FAILURES

Peak afternoon heatwaves overload MSEDCL distribution transformers and municipal water pumping stations, while commercial RO water chillers seize up—shutting off drinking water and cooling at the exact hour thermal vulnerability peaks.

Impact: Static map pins send workers to dry or broken municipal taps.

04 • BLIND NGO DISPATCH & FATAL ANHIDROSIS DELAY

NGO volunteer leads allocate 20L water cans and ORS packets by guesswork rather than mathematical deficit. Meanwhile, victims fail to recognize anhidrosis (cessation of sweating / hot dry skin)—the critical clinical threshold requiring immediate 108 ambulance triage.

Impact: Misallocated relief supplies and delayed emergency medical intervention.

Dual-Portal Spatial Intelligence & Autonomous Relief Dispatch

Connecting outdoor gig workers and NGO volunteer leads through one live geospatial + Gemini agent pipeline.

1. 64-Zone Relief Deficit (Rd) Ranking

Divides Pune (18.41,73.70 to 18.65,73.95) into an 8x8 micro-grid (Z-00 to Z-77). Continuously ranks the top deficit zones over the next 24 hours so NGO leads see where need is highest in < 60 seconds.

2. Multilingual Voice & Text Dispatch Planner

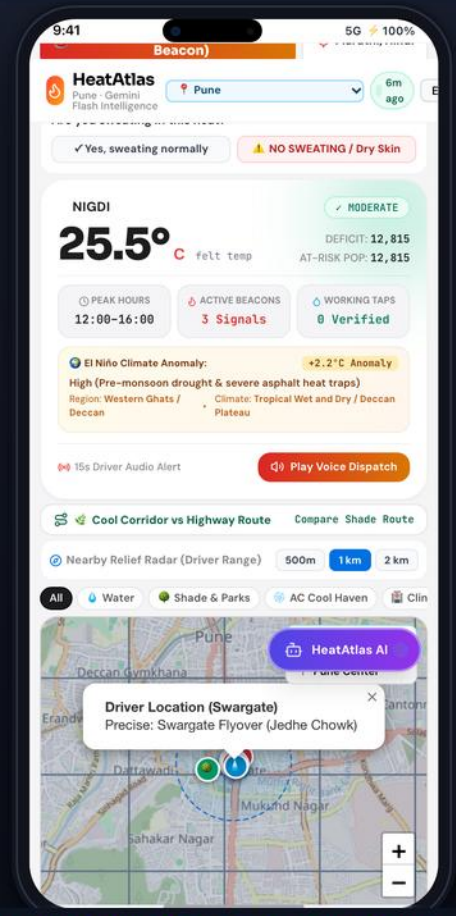
Volunteer leads dictate or type free-form requests in English, Hindi, or Marathi ('Send 2 teams to Yerwada this afternoon, 40 water cans'). Gemini extracts intent, calls live tools, and drafts a WhatsApp brief.

3. Image-First Gemini Vision Anti-Spoofing

Workers report dry/broken municipal taps or crowd heat beacons with a photo. Gemini Vision strictly audits image authenticity first—rejecting spoof/random photos while decrementing working_taps for real failures.

4. 3-Tap Relief Radar & 2-Tap 108 Clinical Triage

Workers find the 3 nearest verified water taps & clinics via Haversine distance in <= 3 taps, or run a 2-tap anhidrosis self-check that triggers emergency 108 routing when sweating stops.



Mobile Worker Portal: Live 64-Zone Map, Relief Radar & Triage

Zero-Hallucination Split: Deterministic Python + Agentic Gemini

Python computes every mathematical score; Gemini 2.5 Flash handles NL extraction, tool routing & verified synthesis.

A. DETERMINISTIC & BIOMETEOROLOGICAL ENGINE (scoring.py + spatial_index.py)

Core Spec Math:	Eh = Pop×hrs(>=38°C) | Vh = (Child+Elderly)×hrs | Rd = Vh / (1 + Taps + 2×Clinics)
WBGT & Canyon SVF:	WBGT (1+0.08×dT^1.35) × SVF Canyon (1+0.15×(1-SVF)) × Saturation (250/tap, 500/clinic)
O(log N) Quadtree:	2D Spatial Quadtree Index for bounding-box range queries & Haversine k-NN relief routing
12 Integrated Repos:	archify • rtk • Prompt_Engineering • refchecker • citefact • draft-detective • remotion • hallmark

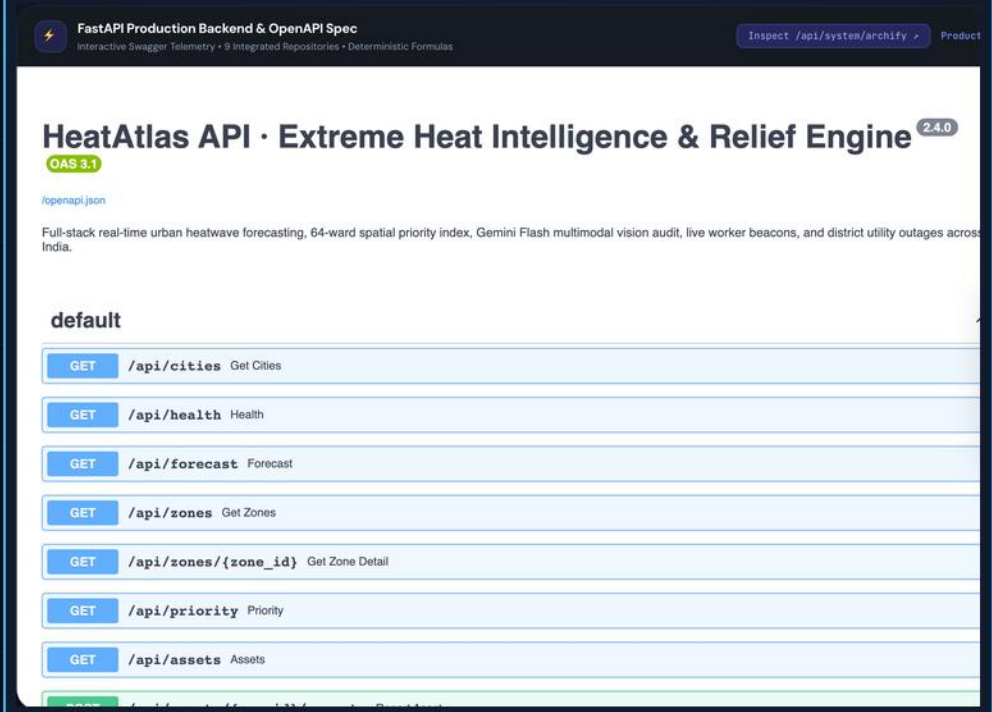
B. NATIVE STRUCTURED OUTPUTS & REFCHECKER GUARDRAILS (agent.py)

1. Native Schema Enforcement: Uses Google GenAI response_schema=DispatchBriefSchema & Pydantic Field(ge=0) models for zero syntax errors.

2. RefChecker + CiteFact Triplet Auditor: Extracts [Zone, has_rd, Value] triplets and verifies 100% of claims against Python ground truth.

3. Automated 108 EMS & WhatsApp Webhooks: /api/webhooks/emergency-108 and /api/webhooks/whatsapp + Offline PWA Service Worker (sw.js).

4. Cross-Platform Vision Anti-Spoofing: Pure in-memory Pillow normalizer + Gemini Vision rejects spoof/non-tap photos automatically.



Production FastAPI OpenAPI 3.1 Spec (/docs) — 12 Live Endpoints

100% Live Public Datasets & FORGE'26 Rubric Alignment

Strict adherence to the Zero-Tolerance Dummy Data Policy — all telemetry streams from live external APIs.

LIVE UPSTREAM TELEMETRY & PERSISTENCE PIPELINE

Open-Meteo Forecast API (Live • 30m TTL)

https://api.open-meteo.com/v1/forecast — Queries 72-hour hourly temperature_2m, relative_humidity_2m, and apparent_temperature across all 64 Pune zone centroids.

WorldPop v2 Demographic API (Live • 24h TTL)

https://api.worldpop.org/v1/services/stats — Extracts total population, under-5 child counts, and 65+ elderly vulnerability rasters per zone.

OpenStreetMap Overpass API (Live • 6h TTL)

https://overpass-api.de/api/interpreter — Indexes live amenity=drinking_water, water_point, clinic, and hospital nodes within [18.41,73.70, 18.65,73.95].

SQLite Crowdsourced State (heatatlas.db)

Persists citizen broken-tap reports with 30-min rate limiting (HTTP 429) and recalculates zone Relief Deficit (Rd) in under 2 seconds.

FORGE'26 JUDGING RUBRIC COVERAGE (100 PTS)

Technical Architecture & Depth (20 Pts)

8x8 geospatial grid indexing, Haversine routing, async httpx clients with retry/cache, SQLite state, and modular FastAPI + React 18 architecture.

AI Reasoning & Agentic Execution (25 Pts)

Multi-step Gemini 2.5 Flash tool-calling loop, structured JSON schema enforcement, Python hallucination verifier, and image-first vision anti-spoofing.

Problem Relevance, Impact & USP (25 Pts)

Replaces city-wide weather averages with hyper-local Relief Deficit (Rd) math, multilingual Marathi/Hindi/EN dispatch, and 2-tap 108 heatstroke triage.

Live Demonstration & Robustness (30 Pts)

Deployed live on Vercel (https://heatatlas.vercel.app/) with 11/11 automated pytest acceptance tests passing + GitHub Actions CI (.github/workflows/ci.yml).

Scaling Autonomous Heat Resilience Across the Global South

Production-ready today in Pune (64 zones), architected for rapid expansion across India's heat-corridor cities.

PHASE 1 — LIVE TODAY (PUNE 64-ZONE GRID)

Full-stack FastAPI + React + Leaflet deployment live at https://heatatlas.vercel.app/. Real-time Open-Meteo, WorldPop & Overpass ingestion, O(log N) Quadtree, RefChecker audit, and 11/11 passing acceptance tests.

PHASE 2 — PAN-INDIA HEAT ACTION PLAN (HAP) ROLLOUT

Pre-configured multi-city bounding boxes for Nagpur, Ahmedabad, Delhi NCR, and Chennai with regional IMD Heat Action Plan threshold calibration and localized voice advisories.

PHASE 3 — MUNICIPAL FLEET & ONDC HYDRATION DISPATCH

Automated webhook integration with municipal water tanker telemetry (PMC / MSEDCL) and ONDC local pharmacy networks for automated ORS & hydration kit restocking in high-deficit wards.

OFFICIAL SUBMISSION SUMMARY

Team Strawhats

TEAM CODE: 627JUH

HACKATHON TRACK

Theme 2: Climate, Resource Infrastructure & Global Supply Resilience

3. PROTOTYPE LINK (LIVE APP)

https://heatatlas.vercel.app/

4. OFFICIAL YOUTUBE DEMO VIDEO

https://youtu.be/C0-BQNND-b8

5. GITHUB REPOSITORY LINK

https://github.com/vishwjeetlol/heatatlas

6. PITCH DECK PDF LINK

https://heatatlas.vercel.app/HeatAtlas_Pitch_Deck.pdf